

Section : Chemistry

Q.1 In alcohols, C-O-H bond angle is

1. Less than the tetrahedral bond angle ($109^{\circ} 28'$)
2. Higher than the tetrahedral bond angle ($109^{\circ} 28'$)
3. Exactly equal to tetrahedral bond angle i.e. $109^{\circ} 28'$
4. Exactly equal to 90°

Options 1. 1

2. 2
3. 3
4. 4

Q.2 Which of the following is used as a catalyst in contact process?

1. finely divided iron
2. V_2O_5
3. Cu_2Cl_2
4. Ni in presence of H_2

Options 1. 1

2. 2
3. 3
4. 4

Q.3 Choose the correct statement about molar conductivity, Λ_m

1. Molar conductivity of a weak electrolyte decreases gradually as the concentration of the electrolyte decreases in the higher concentration ranges
2. Molar conductivity of a strong electrolyte increases linearly with the increase in the concentration of the electrolyte
3. As the concentration of a weak electrolyte approaches zero, there is a sharp increase in the molar conductivity of the solution
4. The limiting molar conductivity of a weak electrolyte can be determined directly from the plot of Λ_m against $c^{1/2}$

Options 1. 1

2. 2
3. 3
4. 4

Q.4 Arrange the following in increasing order of basic strength in aqueous solution :

- (A) $C_6H_5NH_2$
(B) $C_2H_5NH_2$
(C) $(C_2H_5)_2NH$
(D) NH_3

Choose the correct answer from the options given below:

1. (A), (D), (B), (C)
2. (A), (B), (C), (D)
3. (A), (D), (C), (B)
4. (B), (D), (A), (C)

Options 1. 1

2. 2
3. 3
4. 4

Q.5 Which of the following is known as Hinsberg's reagent?

1. Benzenesulphonyl chloride
2. N-Ethylbenzenesulphonamide
3. Diethylbenzenesulphonamide
4. Benzenesulphonyl bromide

Options 1. 1

2. 2
3. 3
4. 4

Q.6 Fuel cell involves _____

1. The reaction of H_2 with O_2 to give H_2O

2. The splitting of H_2O to give O_2
3. The splitting of H_2O to give H_2
4. The reaction of H_2 and H_2O to give O_2

Options 1. 1

2. 2
3. 3
4. 4

Q.7 Choose the correct statement.

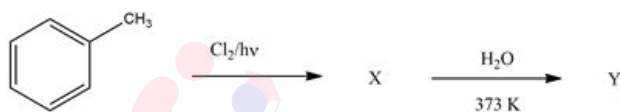
1. Reactions with molecularity three are very rare.
2. Order of reaction cannot have a fraction value.
3. Molecularity of a reaction is meaningful for complex reactions only.
4. Order of a reaction is always determined theoretically

Options 1. 1

2. 2
3. 3
4. 4

Q.8

Identify "Y" in the following reaction:



1. Benzaldehyde
2. 2-Methylbenzaldehyde
3. 2-Chlorophenol
4. Benzal chloride

Options 1. 1

2. 2
3. 3
4. 4

Q.9 Choose the correct statements for a zero order reaction

- (A) The rate of the reaction is independent of the initial concentration of the reactant
- (B) The rate of the reaction increases linearly as the initial concentration of the reactant is increased
- (C) The plot of concentration against time is a straight line with a positive slope
- (D) The unit of rate constant is $\text{mol L}^{-1} \text{s}^{-1}$

Choose the correct answer from the options given below:

1. (A) and (D) only
2. (B), (C) and (D) only
3. (A), (C) and (D) only
4. (B) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.10 Relation between equilibrium constant and standard potential of the cell under standard conditions is represented by :

1. $E_{\text{cell}}^{\circ} = \frac{2.303 RT}{F} \log K_c$
2. $E_{\text{cell}}^{\circ} = \frac{2.303 RT}{nF} \log K_c$
3. $E_{\text{cell}}^{\circ} = \frac{2.303}{nF} \log K_c$
4. $E_{\text{cell}}^{\circ} = \frac{2.303 R}{nF} \log K_c$

Options 1. 1

2. 2
3. 3
4. 4

Q.11 Henry was the first to give a quantitative relation between pressure and solubility of a gas in a solvent. Choose the correct statement given by Henry.

1. The partial pressure of the gas in vapour phase is inversely proportional to the mole fraction of the gas in the solution.}
2. The partial pressure of the gas in vapour phase is proportional to the mole fraction of the gas in the solution.}
3. The total pressure of the gases in vapour phase is proportional to the multiplication of the mole fraction of all the gases present in the solution.
4. The total pressure of the gases in vapour phase is proportional to the sum of the mole fraction of all the gases present in the solution.

Options 1. 1

2. 2
3. 3
4. 4

Q.12 Select the correct statement.

1. For the zero order reaction, the half-life of a reaction is directly proportional to the initial concentration of the reactant, whereas for the first order reaction, the half-life is independent of the initial concentration of the reacting species.
2. For the zero order reaction, the half-life of a reaction is inversely proportional to the initial concentration of the reactant, whereas for the first order reaction, the half-life is independent of the initial concentration of the reacting species.
3. The half-life of a reaction is directly proportional to the initial concentration of the reacting species for both zero and first order reactions.
4. The half-life of a reaction is inversely proportional to the initial concentration of the reacting species for both zero and first order reactions.

Options 1. 1

2. 2
3. 3
4. 4

Q.13 \Large Identify "Y" in the following reaction:



The options for Y are:

1. RCOOH
2. RCHO
3. RCOX
4. ROH

Options 1. 1

2. 2
3. 3
4. 4

Q.14 Arrange the following amines in decreasing order of K_b values in aqueous solution:

- (A) Phenylmethanamine
- (B) N-Ethylethanamine
- (C) N-methylaniline
- (D) Benzenamine

Choose the correct answer from the options given below:

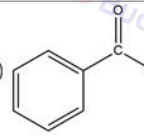
1. (D), (C), (A), (B)
2. (A), (B), (C), (D)
3. (B), (A), (C), (D)
4. (A), (C), (D), (B)

Options 1. 1

2. 2
3. 3
4. 4

Q.15

Match List-I with List-II

List-I	List-II
Name Reaction	Reaction
(A) Stephen Reaction	(I)  $\xrightarrow[\text{Anhyd. AlCl}_3/\text{CuCl}]{\text{CO, HCl}}$ 
(B) Gatterman-Koch Reaction	(II) $\text{RCN} + \text{SnCl}_2 + \text{HCl} \longrightarrow \text{RCH=NH} \xrightarrow{\text{H}_3\text{O}^+} \text{RCHO}$
(C) Etard Reaction	(III)  $\xrightarrow[\text{Pd-BaSO}_4]{\text{H}_2}$ 
(D) Rosenmund reduction	(IV)  + $\text{CrO}_2\text{Cl}_2 \xrightarrow{\text{CS}_2} \text{CH(OCrOHCl}_2)_2 \xrightarrow{\text{H}_3\text{O}^+} \text{CHO}$

Choose the **correct** answer from the options given below:

- (A) - (II), (B) - (I), (C) - (IV), (D) - (III)
- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (II), (B) - (I), (C) - (III), (D) - (IV)
- (A) - (I), (B) - (II), (C) - (IV), (D) - (III)

Options 1. 1

2. 2

3. 3

4. 4

Q.16 Arrange the following alcohols in increasing order of their boiling points.

- (A) Propan-1-ol
(B) Butan-2-ol
(C) Butan-1-ol
(D) Pentan-1-ol

Choose the correct answer from the options given below:

- (A), (B), (C), (D)
- (A), (D), (C), (B)
- (B), (A), (D), (C)
- (C), (B), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.17 Select the correct statement.

- A catalyst does not change the Gibbs energy and equilibrium constant of a reaction.
- A catalyst changes the Gibbs energy but does not change the equilibrium constant of a reaction.
- A catalyst changes the equilibrium constant but does not change the Gibbs energy of a reaction.
- A catalyst changes both the Gibbs energy and equilibrium constant of a reaction

Options 1. 1

2. 2

3. 3

4. 4

Q.18 Match List-I with List-II

List-I	List-II
Compound	Use
(A) Dichloromethane	(I) in the manufacturing of propellants for aerosol cans
(B) DDT	(II) in the production of the Freon refrigerant R-22
(C) Trichloromethane	(III) as a paint remover
(D) Tetrachloromethane	(IV) as an insecticide

Choose the correct answer from the options given below:

- (A) - (II), (B) - (III), (C) - (I), (D) - (IV)
- (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
- (A) - (I), (B) - (III), (C) - (IV), (D) - (II)
- (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.19 The molality of a solution prepared by dissolving 4.0 g NaOH in 1000 mL distilled water is _____
(Assume that the density of water is 1 g mL^{-1})

1. $0.001 \text{ mol kg}^{-1}$
2. 1.0 mol kg^{-1}
3. 0.1 mol L^{-1}
4. 0.1 mol kg^{-1}

Options 1. 1

2. 2

3. 3

4. 4

Q.20 Alcohol is more soluble in water when compared to alkanes due to formation of

1. Covalent bond
2. Ionic bond
3. Hydrogen bond
4. Peptide bond

Options 1. 1

2. 2

3. 3

4. 4

Q.21 Select the correct statements from the following:

- (A) Mn show the highest oxidation state in first transition series elements.
- (B) Titanium (II) is more stable than titanium (III) and titanium (IV).
- (C) Scandium does not exhibit variable oxidation states.
- (D) Zn form complexes with only (II) oxidation state.

Choose the correct answer from the options given below:

1. (A), (B) and (C) only
2. (B), (C) and (D) only
3. (A), (C) and (D) only
4. (A), (B) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.22 Which of the following reactions are given by Phenols?

- (A) Nitration
- (B) Halogenation
- (C) Oxidation
- (D) Acetylation

Choose the correct answer from the options given above:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (B), (C) and (D) only

Options 1. 1

2. 2

3. 3

4. 4

Q.23 Which of the following can be prepared from Sandmeyer's reaction?

- (A) Chlorobenzene
- (B) Bromobenzene
- (C) Iodobenzene
- (D) Fluorobenzene

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (B), (C) and (D)
4. (A) and (B) only

Options 1. 1

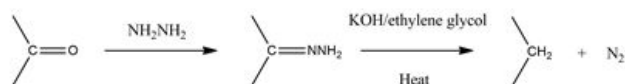
2. 2

3. 3

4. 4

Q.24

Select the correct name for the following reaction:



1. Clemmensen Reduction
2. Wolff-Kishner Reduction
3. Etard Reaction
4. Stephen Reaction

Options 1. 1

2. 2
3. 3
4. 4

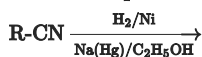
Q.25 Choose the incorrect statement from the following:

1. permanganate ion is tetrahedral
2. pi-bonding in permanganate ion takes place by overlap of p orbital of oxygen with d orbital of manganese
3. permanganate ion is paramagnetic in nature
4. permanganate ion is purple in colour

Options 1. 1

2. 2
3. 3
4. 4

Q.26 What is the product of the following reaction?



The options are:

1. R-CH₂-OH
2. R-CO-NH₂
3. R-NH₂
4. R-CH₂-NH₂

Options 1. 1

2. 2
3. 3
4. 4

Q.27 Match List-I (Name) with List-II (Solution)

List-I

List-II

Name

Solution

- | | |
|-------------------------|---|
| (A) Isotonic solution | (I) A solution with lower osmotic pressure than that of other solution |
| (B) Hypotonic solution | (II) A solution with higher osmotic pressure than that of other solution |
| (C) Ideal solution | (III) Different solutions with equal osmotic pressure |
| (D) Hypertonic solution | (IV) A solution that obeys Raoult's law at all temperatures and concentration |

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (III), (B) - (II), (C) - (IV), (D) - (I)
4. (A) - (III), (B) - (I), (C) - (IV), (D) - (II)

Options 1. 1

2. 2
3. 3
4. 4

Q.28

In the plot of $\log \frac{[R]_0}{[R]}$ versus time for a first order reaction (where [R] represents the concentration of the reactant), the slope and intercept are respectively equal to

1. k and zero
2. k and one
3. $k/2.303$ and zero
4. $2.303k$ and zero

Options 1. 1

2. 2

3. 3

4. 4

Q.29 Arrange the following compounds towards increasing S_N2 reaction?

- (A) n-Butyl bromide
(B) Isobutyl bromide
(C) Tertiary butyl bromide
(D) Sec-Butyl bromide

Choose the correct answer from the options given below:

- (A), (B), (D), (C)
- (A), (B), (C), (D)
- (B), (A), (D), (C)
- (C), (D), (B), (A)

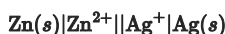
Options 1. 1

2. 2

3. 3

4. 4

Q.30 What is the correct Nernst equation for the following cell?



- $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{RT}{F} \log \frac{[\text{Ag}^+]}{[\text{Zn}^{2+}]}$
- $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{2.303RT}{nF} \log \frac{[\text{Zn}^{2+}]}{[\text{Ag}^+]^2}$
- $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{2.303RT}{F} \log \frac{[\text{Ag}^+]}{[\text{Zn}^{2+}]}$
- $E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{2.303RT}{nF} \log \frac{[\text{Zn}^{2+}]}{[\text{Ag}^+]}$

Options 1. 1

2. 2

3. 3

4. 4

Q.31 Match List-I with List-II

List-I	List-II
Common name	IUPAC name
(A) Vinyl Chloride	(I) 1-Chloro-2-methylbenzene
(B) Chloroform	(II) Dichloromethane
(C) Methylene chloride	(III) Trichloromethane
(D) o-Chlorotoluene	(IV) Chloroethene

Choose the correct answer from the options given below:

- (A) - (I), (B) - (II), (C) - (III), (D) - (IV)
- (A) - (II), (B) - (III), (C) - (IV), (D) - (I)
- (A) - (IV), (B) - (III), (C) - (II), (D) - (I)
- (A) - (IV), (B) - (II), (C) - (III), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.32 Correct electronic configuration of Cu and Cr are _____

- $3d^9 4s^2$ and $3d^4 4s^2$ respectively
- $3d^{10} 4s^1$ and $3d^5 4s^1$ respectively
- $3d^5 4s^1$ and $3d^{10} 4s^1$ respectively
- $3d^4 4s^2$ and $3d^9 4s^2$ respectively

Options 1. 1

2. 2

3. 3

4. 4

Q.33 Which one of the following statements is correct with reference to the solubility of gases in liquid?

- The solubility of gases in liquids increases with the increase of pressure.
- The solubility of gases in liquids is not affected with the increase of temperature.
- The solubility of gases in liquids increases with the increase of temperature.
- At high altitude, the partial vapour pressure of O_2 becomes greater than that at the sea level.

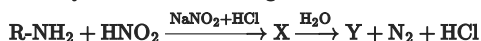
Options 1. 1

2. 2

3. 3

4. 4

Q.34 Identify "Y" in the following reaction:



1. R-CONH₂
2. R-COCl
3. ROH
4. R-NO₂

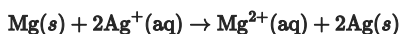
Options 1. 1

2. 2

3. 3

4. 4

Q.35 The standard Gibbs energy of the following reaction is -610 kJ mol^{-1}



What is the standard EMF for the cell corresponding to the above reaction?

Given: Faraday's constant = 96500 C mol^{-1}

1. -6.32 V
2. $+6.32 \text{ V}$
3. $+3.16 \text{ V}$
4. -3.16 V

Options 1. 1

2. 2

3. 3

4. 4

Q.36 Arrange the following compounds in order of their increasing boiling points.

- (A) Pentan-1-ol
- (B) n-Butane
- (C) Pentanal
- (D) Ethoxyethane

Choose the correct answer from the options given below:

1. (A), (B), (D), (C)
2. (B), (D), (C), (A)
3. (A), (C), (D), (B)
4. (B), (C), (D), (A)

Options 1. 1

2. 2

3. 3

4. 4

Q.37 Match List-I with List-II

List-I

Reaction

(A) Hydrolysis of Ethyl Chloride

(B) 2-Propanol is treated with Conc. Sulphuric acid

(C) Reaction of HCN with Acetone

(D) Treatment of acetamide with NaOH and Br₂

List-II

Type of reaction

(I) Substitution reaction

(II) Addition reaction

(III) Degradation reaction

(IV) Elimination reaction

Choose the correct answer from the options given below:

1. (A) - (I), (B) - (IV), (C) - (II), (D) - (III)
2. (A) - (I), (B) - (III), (C) - (II), (D) - (IV)
3. (A) - (I), (B) - (II), (C) - (IV), (D) - (III)
4. (A) - (III), (B) - (IV), (C) - (II), (D) - (I)

Options 1. 1

2. 2

3. 3

4. 4

Q.38 Considering the following statements; identify the correct statements:

- (A) Scandium is regarded as a transition element, but zinc is not.
- (B) V₂O₃ is amphoteric and V₂O₅ is basic oxide.
- (C) The similar radii of Zr and Hf is a consequence of the lanthanoid contraction.
- (D) The oxidation state of chromium in chromate and dichromate ion is same.

Choose the correct answer from the options given below:

1. (A), (B) and (D) only
2. (A), (B) and (C) only
3. (A), (C) and (D) only
4. (B), (C) and (D) only

Options 1. 1

2. 2
3. 3
4. 4

Q.39 Low oxygen concentration in the human body at high altitudes is due to the ____

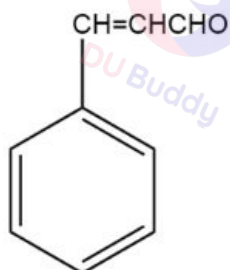
1. Low temperature
2. low atmospheric pressure
3. High atmospheric pressure
4. Both low atmospheric pressure and low temperature

Options 1. 1

2. 2
3. 3
4. 4

Q.40

What is the common name of the given compound?



1. Phthaldehyde
2. Salicylaldehyde
3. Vanillin
4. Cinnamaldehyde

Options 1. 1

2. 2
3. 3
4. 4

Q.41 Read the passage carefully and answer the questions:

The degeneracy of the d orbitals has been removed due to ligand electron-metal electron repulsions in the octahedral complex to yield three orbitals of lower energy, t_{2g} set and two orbitals of higher energy, e_g set. This splitting of the degenerate levels due to the presence of ligands in a definite geometry is termed as crystal field splitting and the energy separation is denoted by Δ_o . Thus energy of the two e_g orbitals will increase by $(3/5)\Delta_o$ and that of three t_{2g} will decrease by $(2/5)\Delta_o$. The crystal field splitting Δ_o depends upon the field produced by the ligand and charge on the metal ion. Some ligands are able to produce strong fields, in which case the splitting will be large, whereas others produce weak fields and consequently result in small splitting of d orbitals. Relative magnitude of crystal field splitting energy Δ_o and pairing energy, P (energy required for electron pairing in a single orbital) determine the formation of low spin ($\Delta_o > P$) or high spin ($\Delta_o < P$) complex. In tetrahedral coordination entity formation, the d -orbital splitting is inverted and is smaller as compared to the octahedral field splitting. The crystal field theory attributes the colour of the complex to $d-d$ transition of the electron. The colour of the coordination compounds depends on the crystal field splitting. In the absence of ligand, crystal field splitting does not occur and hence the substance is colourless.

Which of the following relations is correct for the splitting of d orbitals in octahedral and tetrahedral crystal fields, considering same metal, the same ligands and metal-ligand distances?

1. $\Delta_t = \frac{4}{9} \Delta_o$
2. $\Delta_t = \frac{9}{4} \Delta_o$
3. $\Delta_t = \frac{2}{5} \Delta_o$

$$4. \Delta_t = \frac{3}{5} \Delta_o$$

- Options 1. 1
2. 2
3. 3
4. 4

Q.42 Read the passage carefully and answer the questions:

The degeneracy of the d orbitals has been removed due to ligand electron-metal electron repulsions in the octahedral complex to yield three orbitals of lower energy, t_{2g} set and two orbitals of higher energy, e_g set. This splitting of the degenerate levels due to the presence of ligands in a definite geometry is termed as crystal field splitting and the energy separation is denoted by Δ_o . Thus energy of the two e_g orbitals will increase by $(3/5)\Delta_o$ and that of three t_{2g} will decrease by $(2/5)\Delta_o$. The crystal field splitting Δ_o depends upon the field produced by the ligand and charge on the metal ion. Some ligands are able to produce strong fields, in which case the splitting will be large, whereas others produce weak fields and consequently result in small splitting of d orbitals. Relative magnitude of crystal field splitting energy Δ_o and pairing energy, P (energy required for electron pairing in a single orbital) determine the formation of low spin ($\Delta_o > P$) or high spin ($\Delta_o < P$) complex. In tetrahedral coordination entity formation, the d -orbital splitting is inverted and is smaller as compared to the octahedral field splitting. The crystal field theory attributes the colour of the complex to $d-d$ transition of the electron. The colour of the coordination compounds depends on the crystal field splitting. In the absence of ligand, crystal field splitting does not occur and hence the substance is colourless.

What will be the crystal field splitting energy for tetrahedral $[\text{CoCl}_4]^{2-}$ if its value for octahedral $[\text{CoCl}_6]^{4-}$ is $18,000 \text{ cm}^{-1}$?

1. $40,500 \text{ cm}^{-1}$
2. $10,800 \text{ cm}^{-1}$
3. $20,000 \text{ cm}^{-1}$
4. $8,000 \text{ cm}^{-1}$

- Options 1. 1
2. 2
3. 3
4. 4

Q.43 Read the passage carefully and answer the questions:

The degeneracy of the d orbitals has been removed due to ligand electron-metal electron repulsions in the octahedral complex to yield three orbitals of lower energy, t_{2g} set and two orbitals of higher energy, e_g set. This splitting of the degenerate levels due to the presence of ligands in a definite geometry is termed as crystal field splitting and the energy separation is denoted by Δ_o . Thus energy of the two e_g orbitals will increase by $(3/5)\Delta_o$ and that of three t_{2g} will decrease by $(2/5)\Delta_o$. The crystal field splitting Δ_o depends upon the field produced by the ligand and charge on the metal ion. Some ligands are able to produce strong fields, in which case the splitting will be large, whereas others produce weak fields and consequently result in small splitting of d orbitals. Relative magnitude of crystal field splitting energy Δ_o and pairing energy, P (energy required for electron pairing in a single orbital) determine the formation of low spin ($\Delta_o > P$) or high spin ($\Delta_o < P$) complex. In tetrahedral coordination entity formation, the d -orbital splitting is inverted and is smaller as compared to the octahedral field splitting. The crystal field theory attributes the colour of the complex to $d-d$ transition of the electron. The colour of the coordination compounds depends on the crystal field splitting. In the absence of ligand, crystal field splitting does not occur and hence the substance is colourless.

Choose the correct statement from the following:

1. $\Delta_o > P$; strong field ligand ; low spin complex
2. $\Delta_o > P$; weak field ligand ; low spin complex
3. $\Delta_o < P$; weak field ligand ; low spin complex
4. $\Delta_o < P$; strong field ligand ; high spin complex

- Options 1. 1
2. 2
3. 3
4. 4

Q.44 Read the passage carefully and answer the questions:

The degeneracy of the d orbitals has been removed due to ligand electron-metal electron repulsions in the octahedral complex to yield three orbitals of lower energy, t_{2g} set and two orbitals of higher energy, e_g set. This splitting of the degenerate levels due to the presence of ligands in a definite geometry is termed as crystal field splitting and the energy separation is denoted by Δ_o . Thus energy of the two e_g orbitals will increase by $(3/5)\Delta_o$ and that of three t_{2g} will decrease by $(2/5)\Delta_o$. The crystal field splitting Δ_o depends upon the field produced by the ligand and charge on the metal ion. Some ligands are able to produce strong fields, in which case the splitting will be large, whereas others produce weak fields and consequently result in small splitting of d orbitals. Relative magnitude of crystal field splitting energy Δ_o and pairing energy, P (energy required for electron pairing in a single orbital) determine the formation of low spin ($\Delta_o > P$) or high spin ($\Delta_o < P$) complex. In tetrahedral coordination entity formation, the d -orbital splitting is inverted and is smaller as compared to the octahedral field splitting. The crystal field theory attributes the colour of the complex to $d-d$ transition of the electron. The colour of the coordination

compounds depends on the crystal field splitting. In the absence of ligand, crystal field splitting does not occur and hence the substance is colourless.

Which of the following complex is not expected to be coloured?

1. $[\text{Ti}(\text{H}_2\text{O})_6]^{3+}$
2. $\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
3. $[\text{Zn}(\text{H}_2\text{O})_6]^{2+}$
4. $[\text{Ni}(\text{H}_2\text{O})_6]^{2+}$

Options 1. 1

2. 2

3. 3

4. 4

Q.45 Read the passage carefully and answer the questions:

The degeneracy of the d orbitals has been removed due to ligand electron-metal electron repulsions in the octahedral complex to yield three orbitals of lower energy, t_{2g} set and two orbitals of higher energy, e_g set. This splitting of the degenerate levels due to the presence of ligands in a definite geometry is termed as crystal field splitting and the energy separation is denoted by Δ_o . Thus energy of the two e_g orbitals will increase by $(3/5)\Delta_o$ and that of three t_{2g} will decrease by $(2/5)\Delta_o$. The crystal field splitting Δ_o depends upon the field produced by the ligand and charge on the metal ion. Some ligands are able to produce strong fields, in which case the splitting will be large, whereas others produce weak fields and consequently result in small splitting of d orbitals. Relative magnitude of crystal field splitting energy Δ_o and pairing energy, P (energy required for electron pairing in a single orbital) determine the formation of low spin ($\Delta_o > P$) or high spin ($\Delta_o < P$) complex. In tetrahedral coordination entity formation, the d -orbital splitting is inverted and is smaller as compared to the octahedral field splitting. The crystal field theory attributes the colour of the complex to $d-d$ transition of the electron. The colour of the coordination compounds depends on the crystal field splitting. In the absence of ligand, crystal field splitting does not occur and hence the substance is colourless.

Choose the correct order of the ligands in their increasing crystal field splitting:

1. $\text{Br}^- < \text{NH}_3 < \text{H}_2\text{O} < \text{CO} < \text{CN}^-$
2. $\text{Br}^- < \text{H}_2\text{O} < \text{NH}_3 < \text{CN}^- < \text{CO}$
3. $\text{H}_2\text{O} < \text{NH}_3 < \text{Br}^- < \text{CN}^- < \text{CO}$
4. $\text{CO} < \text{CN}^- < \text{H}_2\text{O} < \text{NH}_3 < \text{Br}^-$

Options 1. 1

2. 2

3. 3

4. 4

Q.46 Read the passage carefully and answer the questions

Optically active polyhydroxy aldehydes or ketones or molecules which provide such units on hydrolysis are called carbohydrates. They can be broadly classified as monosaccharides, disaccharides and polysaccharides depending on their behavior towards hydrolysis. They are also classified as reducing or non-reducing sugars. Another class of biomolecules is proteins which are polymers of twenty different amino acids linked by peptide bonds. Amino acids can be classified as acidic, basic or neutral depending on the relative number of amino and carboxyl groups in their molecule. The essential amino acids cannot be synthesized by our body and hence must be provided through diet. Nucleic acids are the polymers of nucleotides which in turn consist of a base, pentose sugar and phosphate. There are two types of nucleic acids namely, DNA and RNA which differ in the composition of sugar and a base

Which of the following is non-reducing sugar?

1. Maltose
2. Glucose
3. Sucrose
4. Fructose

Options 1. 1

2. 2

3. 3

4. 4

Q.47 Read the passage carefully and answer the questions

Optically active polyhydroxy aldehydes or ketones or molecules which provide such units on hydrolysis are called carbohydrates. They can be broadly classified as monosaccharides, disaccharides and polysaccharides depending on their behavior towards hydrolysis. They are also classified as reducing or non-reducing sugars. Another class of biomolecules is proteins which are polymers of twenty different amino acids linked by peptide bonds. Amino acids can be classified as acidic, basic or neutral depending on the relative number of amino and carboxyl groups in their molecule. The essential amino acids cannot be synthesized by our body and hence must be provided through diet. Nucleic acids are the polymers of nucleotides which in turn consist of a base, pentose sugar and phosphate. There are two types of nucleic acids namely, DNA and RNA which differ in the composition of sugar and a base

Four bases present in RNA molecule are

1. Adenine, Guanine, Cytosine, Thymine
2. Adenine, Guanine, Thymine, Uracil

3. Guanine, Cytosine, Thymine, Uracil
4. Adenine, Guanine, Cytosine, Uracil

- Options 1. 1
2. 2
3. 3
4. 4

Q.48 Read the passage carefully and answer the questions

Optically active polyhydroxy aldehydes or ketones or molecules which provide such units on hydrolysis are called carbohydrates. They can be broadly classified as monosaccharides, disaccharides and polysaccharides depending on their behavior towards hydrolysis. They are also classified as reducing or non-reducing sugars. Another class of biomolecules is proteins which are polymers of twenty different amino acids linked by peptide bonds. Amino acids can be classified as acidic, basic or neutral depending on the relative number of amino and carboxyl groups in their molecule. The essential amino acids cannot be synthesized by our body and hence must be provided through diet. Nucleic acids are the polymers of nucleotides which in turn consist of a base, pentose sugar and phosphate. There are two types of nucleic acids namely, DNA and RNA which differ in the composition of sugar and a base

"Maltose" disaccharide is composed of

1. Two α -D-glucose units
2. α -D-glucose and β -D-fructose
3. β -D-galactose and β -D-glucose
4. α -D-glucose and β -D-glucose

- Options 1. 1
2. 2
3. 3
4. 4

Q.49 Read the passage carefully and answer the questions

Optically active polyhydroxy aldehydes or ketones or molecules which provide such units on hydrolysis are called carbohydrates. They can be broadly classified as monosaccharides, disaccharides and polysaccharides depending on their behavior towards hydrolysis. They are also classified as reducing or non-reducing sugars. Another class of biomolecules is proteins which are polymers of twenty different amino acids linked by peptide bonds. Amino acids can be classified as acidic, basic or neutral depending on the relative number of amino and carboxyl groups in their molecule. The essential amino acids cannot be synthesized by our body and hence must be provided through diet. Nucleic acids are the polymers of nucleotides which in turn consist of a base, pentose sugar and phosphate. There are two types of nucleic acids namely, DNA and RNA which differ in the composition of sugar and a base

The sugar moiety in DNA and RNA molecules, respectively, are

1. β -D-ribose and β -D-3-deoxyribose
2. β -D-2-deoxyribose and β -D-ribose
3. β -D-ribose and β -D-2-deoxyribose
4. β -D-3-deoxyribose and β -D-ribose

- Options 1. 1
2. 2
3. 3
4. 4

Q.50 Read the passage carefully and answer the questions

Optically active polyhydroxy aldehydes or ketones or molecules which provide such units on hydrolysis are called carbohydrates. They can be broadly classified as monosaccharides, disaccharides and polysaccharides depending on their behavior towards hydrolysis. They are also classified as reducing or non-reducing sugars. Another class of biomolecules is proteins which are polymers of twenty different amino acids linked by peptide bonds. Amino acids can be classified as acidic, basic or neutral depending on the relative number of amino and carboxyl groups in their molecule. The essential amino acids cannot be synthesized by our body and hence must be provided through diet. Nucleic acids are the polymers of nucleotides which in turn consist of a base, pentose sugar and phosphate. There are two types of nucleic acids namely, DNA and RNA which differ in the composition of sugar and a base

The side chain (R group) present in the "Aspartic Acid" amino acid is

1. $\text{H}_3\text{C-S-CH}_2\text{-CH}_2\text{-}$
2. $\text{HOOC-CH}_2\text{-}$
3. $\text{HOOC-CH}_2\text{-CH}_2\text{-}$
4. $\text{HO-CH}_2\text{-}$

- Options 1. 1
2. 2
3. 3
4. 4